

Rintaro Masaoka

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Research interests

I am interested in exploring universal properties of quantum phases of matter and quantum phase transitions. My current research focuses on frustration-free quantum systems, which exhibit distinctive properties in both gapped and gapless phases. My main research directions are:

- Proving a universal inequality on dynamical exponents for frustration-free quantum critical points.
- Formulating and characterizing frustration-free free-fermion systems.

Education and Positions

September 2026 - Present, Ph.D. in Physics, Watanabe Group, Department of Physics, The Hong Kong University of Science and Technology

April 2026 - August 2026, Research Assistant, Watanabe Group, Department of Physics, The Hong Kong University of Science and Technology

April 2024 - March 2026, Master of Engineering in Applied Physics, Watanabe Group, Department of Applied Physics, Graduate School of Engineering, The University of Tokyo

April 2020 - March 2024, Bachelor of Science in Physics, Department of Physics, Faculty of Science, The University of Tokyo

Awards

2026, Outstanding Presentation Award (Parallel Session I), 71st Condensed Matter Physics Summer School

2026, Dean's Award, Graduate School of Engineering, The University of Tokyo

2026, Shoji Tanaka Award (Outstanding Master's Thesis in the Department of Applied Physics)

2025, Selected as a JSPS Research Fellow (DC1, FY2026; declined)

2024, Outstanding Presentation Award (Parallel Session, Statistical Mechanics), 69th Condensed Matter Physics Summer School

Fellowships and Funding

Hong Kong PhD Fellowship Scheme (HKPFS), Research Grants Council, Hong Kong (2026 - 2029)

Research Assistantship, WINGS Program "Forefront Physics and Mathematics Program to Drive Transformation" (FoPM) (October 2024 - February 2026)

Publications

Preprints

1. **Rintaro Masaoka**. “ $c = -2$ conformal field theory in quadratic band touching.” *arXiv:2511.16496*.

Journal Articles

1. **Rintaro Masaoka**, Seishiro Ono, Hoi Chun Po, Haruki Watanabe. “Frustration-free fermionic systems: Finite-size scaling and band topology.” *Phys. Rev. B* **113**, 195120 (2026).
2. Seishiro Ono, **Rintaro Masaoka**, Haruki Watanabe, Hoi Chun Po. “Frustration-free free fermions.” *Phys. Rev. Research* **8**, 023171 (2026).
3. **Rintaro Masaoka**, Tomohiro Soejima, Haruki Watanabe. “Rigorous lower bound on dynamic critical exponents in critical frustration-free systems.” *Phys. Rev. X* **15**, 041050 (2025).
4. **Rintaro Masaoka**, Tomohiro Soejima, Haruki Watanabe. “Rigorous Lower Bound of the Dynamical Critical Exponent of the Ising Model.” *J. Stat. Phys.* **192**, 76 (2025).
5. **Rintaro Masaoka**, Tomohiro Soejima, Haruki Watanabe. “Quadratic dispersion relations in gapless frustration-free systems.” *Phys. Rev. B* **110**, 195140 (2024).